

1. Architecture & Design Hotspots Analysis

Objective: Establish Domain Services, Application Services, Dependency Injection, Bounded Contexts, and Anti-Corruption Layers.

Date: 2026-07-23 | **Scope:** multi-agent-web-ui — React + Node.js/Express + Vendor Microservices

Executive Summary

EXECUTIVE SUMMARY

The codebase exhibits significant architectural debt across both frontend and backend layers. Critical issues include oversized components (17 React components >400 LOC, with useAppStore.jsx at 3800 LOC), missing service layer abstraction in the frontend where API calls are scattered across components, and a monolithic state management approach that violates separation of concerns. The backend demonstrates better modularity with thin route handlers, but lacks formal repository patterns and has some circular dependency risks between lib modules and components.

32

CONTROLLERS / HANDLERS

4

MODELS / ENTITIES

53

SERVICE CLASSES FOUND

0

REPOSITORY CLASSES
FOUND

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by High-Risk God Classes/Components and Missing Frontend Service Layer.

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	93	GOOD
H2	Missing Service Layer	Controllers accessing repos/models	<10	10–20	>20	8	GOOD
H3	Missing Repository Pattern	Direct DB access points	<10	10–20	>20	6	GOOD
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	2	MODERATE

H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	3	MODERATE
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100%	GOOD
H7	God Classes	Classes >1000 LOC	0	1–3	>3	5	HIGH RISK
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	2	GOOD
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	5%	GOOD
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	381	HIGH RISK
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	25	HIGH RISK
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	17	HIGH RISK
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	5+ levels	HIGH RISK
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	1	GOOD